

HD26

User Manual

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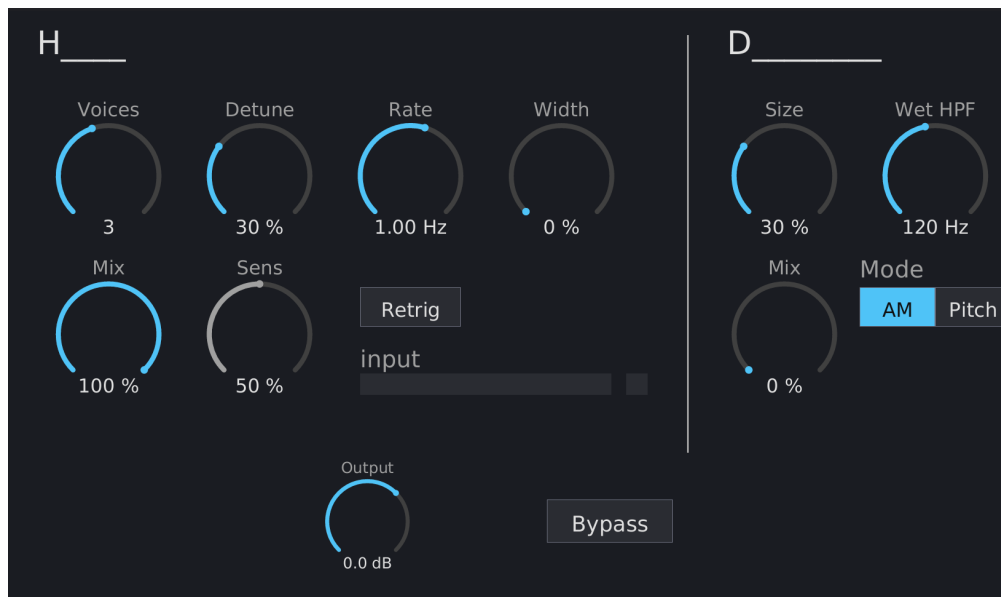


Figure 1: HD26 interface

What is HD26?

HD26 is a stereo “thicken and widen” effect built from two independent sections wired in series. On the panel they are labelled **H**___ (left) and **D**_____ (right):

- **H**___ is a multi-voice **unison-detune chorus**. It generates up to seven detuned copies of the input whose pitch gently oscillates sharp and flat, turning a thin or mono source into a thick, ensemble-like texture — the same kind of “supersaw” thickening you’d get from oscillator unison, but applied to any audio. It can optionally re-fan its detune from centre on each detected transient.
- **D**_____ is a **pseudo-stereo widener**. It adds perceived stereo width to a mono or narrow source using a bank of short, out-of-phase delays. Its default mode is **mono-compatible**: when the mix is folded to mono, the widening cancels and you get the dry signal back.

Both sections have their own **Mix** control and can be used alone or together. The whole chain is feed-forward, so HD26 reports **zero latency** and uses no FFT — it is light enough to run on many tracks at once.

Installation

Build from source (requires stable Rust):

```
cargo nih-plugin bundle hd26 --release
```

The bundler outputs to target/bundled/. Copy either the .vst3 or .clap file to your plugin directory:

- **Linux:** ~/.vst3/ or ~/.clap/
- **macOS:** ~/Library/Audio/Plug-Ins/VST3/ or ~/Library/Audio/Plug-Ins/CLAP/
- **Windows:** C:\Program Files\Common Files\VST3\ or C:\Program Files\Common Files\CLAP\

A standalone build is also available (cargo build --bin hd26 --release), useful for testing the GUI without a host.

Quick Start

1. Insert HD26 on a track and play audio through it.
2. **H___** is active by default (**Voices** = 3, **Mix** = 100 %), so you'll hear thickening immediately. Raise **Detune** for a wider spread and **Rate** to make the movement faster.
3. **D_____** starts silent (**Mix** = 0 %). Raise its **Mix** to add stereo width. Leave **Mode** on **AM** for mono-safe widening, or switch to **Pitch** for a lush, chorus-like motion.
4. To make the detune re-fan rhythmically on each note or hit, turn on **Retrig** and adjust **Sens** while watching the activity LED.
5. Trim the final level with **Output**.

Controls

HD26's signal flows **input** → **H___** → **D_____** → **Output**. Each section is blended back against the dry signal by its own **Mix** knob, so you can dial in as much or as little of each as you like.

In your host's parameter list the controls are prefixed to show which section they belong to — H Detune, H Mix, D Size, D Mix, and so on — since both sections share some control names (notably **Mix**).

H___ — unison-detune chorus (left section)

Voices Number of detuned voices, **0 - 7** (default **3**).

Each voice is a pitch-shifted copy of the input. More voices = a thicker, denser ensemble. **At 0 the section is bypassed** and passes the dry signal through untouched.

Detune Spread depth, **0 - 100 %** (default **30 %**).

Sets how far apart the voices are detuned. The response is deliberately gentle at low settings and opens up steeply near the top: small amounts give a subtle chorus shimmer, high amounts give a wide, supersaw-style fan. The voices are spread asymmetrically around the centre pitch (an even fan of sharp and flat copies), and as more voices are added the outer ones reach further.

Rate Movement speed, **0.01 - 10 Hz** (default **1 Hz**).

The voices' pitch continuously oscillates sharp and flat; Rate is the speed of that oscillation. Low rates give a slow, evolving swell; higher rates give a faster, more obvious chorus warble.

Each voice runs at a slightly different rate so the movement never lines up into a single audible wobble.

Width Stereo spread of the ensemble, **0 - 100 %** (default **0 %**).

At **0 %** the left and right channels are processed identically, so a mono input stays perfectly mono (fully mono-safe). Turning Width up decorrelates the two channels — the voices' movement is offset between left and right — spreading the ensemble across the stereo field. (Width on this section is independent of the D_____ widener; you can use either or both.)

Mix Dry/wet for the H___ section, **0 - 100 %** (default **100 %**).

Blends the detuned ensemble against the dry signal. At 0 % only the dry signal passes; at 100 % you hear the full ensemble.

Sens Transient sensitivity for **Retrig**, **0 - 100 %** (default **50 %**).

Higher values make the detector fire on smaller transients (more re-triggers); lower values respond only to strong hits. The detector is level-independent — it reacts to the *relative* jump in level, not the absolute loudness — so it behaves the same on quiet and loud material.

Because Sens only affects the audio when **Retrig** is on, the dial is **dimmed when Retrig is off** to show it isn't currently acting on the sound. It remains fully adjustable, and the activity LED stays live (see below) so you can dial Sens in before enabling Retrig.

Retrig Transient re-trigger, **on / off** (default **off**).

When on, HD26 watches the incoming audio for transients (note onsets, drum hits). On each detected transient it **re-aligns every voice oscillation to its starting phase**, snapping the ensemble back to a consistent state — a short “zap” that re-articulates the detune on every hit. This is driven entirely by the audio; no MIDI is required. Typical chorus/ensemble use leaves this off; it is most useful on percussive or monophonic material.

Activity meter (input bar + LED) Below the Retrig button is a small activity display:

- The **input bar** shows the level the transient detector is seeing.
- The **LED** flashes on every detected transient.

The LED flashes whenever a transient is detected, **even when Retrig is off**, so you can watch it and set **Sens** before turning Retrig on. When Retrig is on, each flash also corresponds to a voice re-trigger.

D_____ — pseudo-stereo widener (right section)

Size Delay amount, **0 - 100 %** (default **30 %**).

Scales the length of the widener's internal delays (from roughly a millisecond up to around twenty). Larger sizes produce more perceived width, with more time-smearing of transients; smaller sizes are tighter and more subtle.

Wet HPF High-pass on the widener's wet signal, **20 - 500 Hz** (default **120 Hz**).

Keeps low frequencies out of the widened (side) signal so the bass stays centred and solid. Raise it if low end feels loose or smeared on mono systems; lower it to let more of the low range be widened.

Mix Dry/wet for the D_____ section, **0 - 100 %** (default **0 %**).

At 0 % the section is disabled — on a fresh insert the widener is silent until you raise this. Increase it to blend in the widened signal.

Mode Widening character: **AM** or **Pitch** (default **AM**).

- **AM** (default) — the widener’s delay taps are *amplitude*-modulated and summed out of phase, injecting the wet signal purely into the stereo “sides”. This is **mono-compatible**: if the output is folded to mono, the widening cancels and the dry signal returns unchanged. There is no pitch movement, only a gentle, slowly-shifting width.
- **Pitch** — the delay *times* are modulated, with the channels cross-fed in opposite polarity. This gives a richer, more classic chorus-style motion with audible pitch movement, at the cost of full mono-compatibility (some of the effect will partially cancel when summed to mono).

Choose **AM** when mono fold-down matters (broadcast, club systems); choose **Pitch** when you want a lush, more obviously moving stereo image.

Global

Output Output trim, **−24 ... +12 dB** (default **0 dB**).

Final level after both sections. Useful because high Detune/Voices or a wide D_____ setting can change the perceived loudness.

Bypass Plugin bypass, **on / off** (default **off**).

When on, the input passes straight through, unprocessed.

Interacting with the GUI

- **Drag** a dial up/down to change its value; hold **Shift** while dragging for fine adjustment.
- **Double-click** a dial to reset it to its default.
- **Right-click** a dial to type an exact value (Enter to commit, Escape to cancel).
- Click the **Retrig** / **Bypass** buttons and the **Mode** selector to toggle them.
- The window is **freely resizable**, and its size is saved with your project.

How it works

H____ is a multi-voice modulated-delay chorus. Each voice reads a shared, per-channel delay line at a position that is swept by its own low-frequency oscillator; the moving read position Doppler-shifts the pitch up and down, producing the sharp/flat oscillation. The voices’ detune amounts are spread asymmetrically around the centre (an even fan of sharp and flat copies), scaled by **Detune**, with **Rate** setting the oscillation speed. **Width** offsets the left/right oscillation phase to decorrelate the channels. **Retrig** resets every voice oscillator to its initial, evenly-staggered phase on a detected transient, re-synchronising the ensemble.

D_____ in **AM** mode is a four-tap complementary-comb widener: four short delays taken from the mono sum, each scaled by a slow amplitude oscillator and summed out of phase, then injected as a pure side signal (+ to left, − to right). Because the wet is purely “side”, summing left + right returns the dry mono exactly — that is what makes it mono-compatible. **Pitch** mode instead modulates the delay times and cross-feeds the channels in opposite polarity for a more vibrato-like character.

The transient detector that drives **Retrig** compares a fast envelope to a slow one and fires when their ratio crosses the **Sens**-derived threshold, with a short refractory hold to avoid double-triggering. Because it works on the ratio, it is independent of input level.

The whole chain is feed-forward (a dry path plus delayed/processed taps), so HD26 reports **zero latency** and performs no FFT.

Parameter summary

Section	Control	Range	Default
H___	Voices	0 - 7	3
H___	Detune	0 - 100 %	30 %
H___	Rate	0.01 - 10 Hz	1 Hz
H___	Width	0 - 100 %	0 %
H___	Mix	0 - 100 %	100 %
H___	Sens	0 - 100 %	50 %
H___	Retrig	off / on	off
D_____	Size	0 - 100 %	30 %
D_____	Wet HPF	20 - 500 Hz	120 Hz
D_____	Mix	0 - 100 %	0 %
D_____	Mode	AM / Pitch	AM
Global	Output	-24 ... +12 dB	0 dB
Global	Bypass	off / on	off

Tips

- **Thicken a mono lead or bass:** H___ with Voices 3-7 and a moderate Detune; leave Width at 0 to keep it mono-safe, then add stereo with D_____ in AM mode.
- **Wide pad:** push H___ Detune and Width up, then add D_____ (try Pitch mode) for a lush, moving image.
- **Stay mono-safe:** prefer D_____ **AM** mode and keep H___ **Width** modest; the **Wet HPF** keeps the low end centred.
- **Rhythmic re-fan:** on drums or plucks, enable **Retrig** and tune **Sens** by watching the LED so the detune snaps back on each hit.
- **Both sections have a Mix:** if you only want one effect, leave the other section's Mix at 0 % (D_____ is already at 0 % by default; set Voices to 0 to fully bypass H___).